
Seasonal Variation in Spatial Extremal Dependence of Wind Gusts in the Netherlands

Supervisor I: dr. Yicong Lin

Supervisor II: dr. Wiebke Jaeger (Institute for Environmental Studies)

Name	Student Number	Email
F.S. de Jong	2857580	f.s.de.jong2@student.vu.nl

1 Introduction

Damaging wind in the Netherlands arrives in two distinct forms. Winter gusts are produced by extra-tropical cyclones, whose mature wind fields can cover large parts of the country within a few hours, so that many places are extreme at once. Summer gusts are more often produced by convective storms and local downbursts, which are intense but spatially confined, so that one station can record a severe gust while nearby stations do not (mohr2017; gliksman2023european). For the magnitude of the gust at a single site these may look alike; for the spatial extent of the event they differ, and that extent determines how much of a network is stressed at the same time. The object of this thesis is the spatial extent of wind-gust extremes, and whether it differs between the two seasons.

Spatial extent is a question about dependence in the joint tail, not about marginal magnitudes, and ordinary correlation does not answer it: correlation is dominated by ordinary days and need not describe how the most extreme gusts co-occur. The relevant quantity is extremal dependence, which describes how the probability that two sites are jointly extreme behaves as the level rises and as the distance between them grows. Estimating it requires extreme-value methods rather than second-moment summaries, and estimating it as a function of distance requires a spatial model for the tail.

Max-stable processes provide such a model: they are the limiting fields for spatial maxima, just as the generalised extreme value law is the limit for scalar maxima (Haan and Ferreira, 2006; Davison, Padoan, et al., 2012). Within this class the Brown–Resnick process (Brown and Resnick, 1977; Kabluchko et al., 2009) is the model used here. Its extremal dependence is set by the variogram of an underlying Gaussian process, taken as $\gamma_c(d) = (d/\rho_c)^{\alpha_c}$, in which the range ρ_c controls how slowly dependence decays with distance — a larger ρ_c is a larger spatial footprint of extremes — and α_c controls the smoothness of that decay. The process is used as a parametric approximation to the season-specific dependence structure, not as a claim that daily wind gusts are exactly max-stable. Formulating the question through ρ turns a comparison of spatial footprints into a comparison of one interpretable parameter between seasons.

The model cannot be fitted by full likelihood once more than a few stations are involved, because the joint density of a max-stable field is intractable in moderate dimension (Padoan et al., 2010). The estimator is therefore the pairwise composite likelihood of Padoan et al. (2010): the intractable joint density is replaced by a sum of bivariate densities over station pairs, which is estimable on the KNMI network and is the same composite-likelihood device used elsewhere in econometrics to replace an infeasible likelihood by a tractable objective. Daily maxima are treated as repeated spatial fields, their margins are standardised to a common unit-Fréchet scale by station and season, and the Brown–Resnick parameters are estimated separately for winter and summer.

1.1 Research question

Does the spatial range of extremal dependence in Dutch daily maximum wind gusts differ between winter and summer?

Writing $\vartheta_c = (\rho_c, \alpha_c)$ for the season-specific Brown–Resnick parameters, the comparison is of the dependence ranges. The expectation that winter storms have broader footprints than summer convection is the one-sided hypothesis

$$H_0 : \rho_W = \rho_S \quad \text{against} \quad H_1 : \rho_W > \rho_S.$$

The comparison is thus of estimated parameters and their uncertainty, not only of empirical curves; the implied extremal-coefficient curves $\theta_c(d)$ are reported to place the parameter difference on the dependence-versus-distance scale.

1.2 Contribution and outline

The contribution is to estimate and validate a season-specific parametric spatial extremes model for Dutch wind gusts. Brown–Resnick dependence parameters are estimated separately for winter and summer by pairwise composite likelihood, the range difference $\rho_W - \rho_S$ is tested by a seasonal-year block bootstrap, and a Monte Carlo study under the observed station geometry establishes whether the estimator and the test recover a known seasonal difference before the same procedure is applied to the data. The nonparametric tail-dependence coefficient, F -madogram and empirical extremal coefficient, which are standard nonparametric summaries of extremal dependence, are retained as diagnostics for the fitted model.

Section 2 reviews extreme-value theory, the Brown–Resnick model and composite-likelihood inference. Section 3 defines the estimator and the seasonal comparison. Section 4 reports the simulation. Section ?? describes the data and Section ?? the empirical estimates, and Section ?? concludes.

2 Literature Review

This section reviews the literature on extreme-value theory for environmental data, spatial extremes and max-stable processes, pairwise extremal-dependence measures, pairwise extremal-dependence measures, simulation-based validation, uncertainty for dependence estimators, and seasonal variation in extremal dependence.

2.1 Extreme-value theory

Extreme-value theory provides the statistical framework for studying rare events in the tails of a distribution. For the canonical case of maxima, let $X_1, \dots, X_n$ be independent observations with common distribution function F and let $M_n = \max\{X_1, \dots, X_n\}$. Then

$$\Pr\{M_n \leq z\} = \Pr\{X_1 \leq z, \dots, X_n \leq z\} = F(z)^n.$$

The results of Fisher and Tippett (1928) and Gnedenko (1943) show that, under normalisation, any non-degenerate limit of normalised sample maxima must belong to the generalised extreme value (GEV) family, of which the Gumbel, Fréchet, and Weibull types are special cases. This justifies modelling tail observations directly and provides the basis for the block-maxima approach, in which the data are divided into non-overlapping blocks and the maximum within each block is modelled. Block maxima are also the entry point for spatial extremes, since spatial max-stable processes arise as the limiting objects for spatial block maxima (Haan and Ferreira, 2006).

In our paper, hourly observations are first aggregated to daily maximum wind gusts. These daily maxima form the empirical time series from which seasonal high-threshold exceedances are selected. Daily maxima sit between two extremes of block length. Raw hourly gusts are strongly serially dependent, as consecutive hours typically belong to the same storm system, whereas seasonal maxima would leave only about 35 observations per station per season over 1991–2025, too few to estimate pairwise dependence stably. Daily maxima retain on the order of ninety observations per station per season while reducing the most immediate within-day clustering.

An alternative is the peaks-over-threshold (POT) framework, in which exceedances above a high threshold are approximated by a generalised Pareto distribution (**balkema1974; pickands1975**). For hourly data, consecutive exceedances usually belong to the same storm and require declustering (**ferro2003**) or explicit temporal modelling (**fawcett2006**). Daily aggregation already addresses the most immediate serial dependence as we will use the augmented dickey-fuller test (**DickeyFuller1979**). The empirical extremes are defined through a peaks-over-threshold logic applied to the daily maximum series. For each station and season, the main threshold is the empirical 99th percentile. Thus, an extreme event at station i in season c is an observation satisfying

$$X_{i,t} > q_{i,c}(0.99),$$

where $q_{i,c}(0.99)$ denotes the station- and season-specific 99th percentile. This makes the winter and summer samples comparable in tail probability rather than in absolute wind speed level. Because the 99th percentile leaves relatively few exceedances, lower high thresholds such as 95 and 97.5 percent are used as robustness checks.

The marginal step is only the first part of the problem. The spatial question is: conditional on one station observing an extreme daily maximum wind gust, how likely is another station to observe an extreme value on the same day? This shifts the focus to spatial extremal dependence, whose estimands are pairwise tail-dependence measures, extremal coefficients, and dependence–distance curves.

2.2 Spatial extremes and max-stable processes

For block maxima observed over space, max-stable processes provide the standard limiting framework, arising as the limit of pointwise maxima of independent replications of a spatial random field, just as the GEV distribution arises as the limit of scalar maxima. Extremal dependence thereby becomes a spatial object, described as a function of location and distance. Haan and Ferreira (2006) give a foundational

treatment of stationary max-stable models, showing how spatial extremal dependence can be governed by a few parameters related to inter-station distance. Their stationarity assumption imposes a homogeneous dependence structure, which may be restrictive for Dutch gusts, since winter and summer storms need not share the same spatial pattern.

Davison, Padoan, et al. (2012) review the main approaches to spatial extremes and conclude that good marginal modelling is not sufficient: models that fit univariate extremes well may still misrepresent joint extremes across locations. This motivates our focus on extremal-dependence summaries rather than marginal behaviour alone. Full-likelihood inference for max-stable processes is infeasible once the number of locations is moderate, and Padoan et al. (2010) replace it with a pairwise composite likelihood, establishing pairwise inference as a standard approach.

In our paper, max-stable processes serve mainly as a theoretical reference point and as the benchmark data-generating process for the simulation study (Section 2.4), not as a model fitted to the data. We adopt the pairwise logic of Padoan et al. (2010), but the empirical objects are nonparametric pairwise summaries computed directly from the seasonal daily maximum wind gusts.

2.3 Pairwise extremal-dependence measures

The analysis requires dependence measures interpretable at the station-pair level. **cooley2012** survey such tools, including extremal coefficients, madograms, and tail-dependence summaries, which reduce the high-dimensional dependence problem to pairwise quantities that can be compared across seasons and plotted against distance. In our paper, these measures are estimated from station- and season-standardised daily maximum wind gusts, separately for winter (DJF) and summer (JJA), with margins transformed separately by station and season so that the comparison reflects differences in spatial dependence rather than in the marginal distributions.

For two locations s_i and s_j , the pairwise extremal coefficient $\theta(s_i, s_j)$ lies in $[1, 2]$: values close to one indicate strong extremal dependence and values close to two near-independence, so stronger winter dependence corresponds to lower winter $\theta(s_i, s_j)$ at comparable distances. An empirical route to it is the F -madogram. Cooley et al. (2006) study madogram-based summaries for max-stable random fields; for transformed margins,

$$\nu_F(s_i, s_j) = \frac{1}{2} \mathbb{E} [|F_{s_i}\{X(s_i)\} - F_{s_j}\{X(s_j)\}|],$$

where F_{s_i} and F_{s_j} are the marginal distribution functions, and under max-stability

$$\theta(s_i, s_j) = \frac{1 + 2\nu_F(s_i, s_j)}{1 - 2\nu_F(s_i, s_j)}.$$

The F -madogram thus gives a direct nonparametric estimate of a dependence measure summarised as a dependence–distance curve. The upper-tail dependence coefficient χ gives a complementary view. With $U(s_i)$ and $U(s_j)$ the marginally standardised observations on a uniform scale,

$$\chi(s_i, s_j) = \lim_{u \uparrow 1} \Pr\{U(s_j) > u \mid U(s_i) > u\},$$

and the finite-level quantity is

$$\chi_u(s_i, s_j) = \Pr\{U(s_j) > u \mid U(s_i) > u\}.$$

The main empirical specification sets $u = 0.99$, corresponding to co-exceedance of the seasonal upper one percent of daily maximum wind gusts. Sensitivity to $u = 0.95$ and $u = 0.975$ is reported to check whether the results depend on a single threshold choice. Computed within each season from daily co-exceedances, χ_u answers the research question directly: if one KNMI station records an extreme daily maximum wind gust, how likely is another to do so on the same day? Larger χ_u indicates stronger finite-level co-exceedance, just as smaller θ indicates stronger extremal dependence. Coles et al. (1999) introduce χ and $\bar{\chi}$ to distinguish asymptotic dependence from asymptotic independence and discuss their finite-level estimation. **wadsworth2012** and **huser2022** caution that classical max-stable models are natural for asymptotic dependence but that finite data may show weakening dependence at increasingly high levels; our paper therefore reports finite-level dependence on a small grid of thresholds rather than a single threshold.

2.4 Data-generating processes for the simulation study

To check whether the estimators recover a true dependence structure, recovery must be measured against a known dependence–distance relation. The pairwise extremal coefficients of max-stable processes are available in closed form, so a simulated field carries a known $\theta(s_i, s_j)$ at every distance against which the estimated curve can be compared. The Brown–Resnick process (**brown1983**), characterised through a negative-definite variogram, is a widely used family whose extremal dependence is controlled by that variogram and ranges from strong to near-independent as its range varies, and Dombry et al. (2016) provide exact simulation algorithms that ensure the known dependence structure is the one actually simulated.

Our paper uses the Brown–Resnick process as the benchmark data-generating process, since its variogram gives direct control over how extremal dependence decays with distance. The null and alternative designs set its variogram-range parameters so that the two regimes either share a dependence structure or differ by a controlled amount. The Brown–Resnick process is used only as a controlled benchmark whose true dependence is known; we do not assume that the Dutch wind-gust data are Brown–Resnick. The aim is solely to test whether the pairwise estimators recover a known winter-versus-summer difference in spatial extremal dependence.

2.5 Inference and uncertainty for dependence estimators

Because our paper compares estimated dependence–distance curves, a difference between two curves is only informative once the uncertainty in each is quantified. Two sources of variability matter. First, the F -madogram and χ_u estimators are empirical means and conditional frequencies computed from a finite number of high-threshold daily maxima, so their variance grows as the season-split sample shrinks. Second, although daily aggregation removes within-day clustering, a multi-day storm can produce extreme daily maxima on consecutive days, so daily observations cannot be treated as fully independent.

Our paper therefore quantifies uncertainty by resampling temporal blocks. A year or seasonal-block bootstrap resamples whole years, or whole seasonal segments, with replacement and recomputes the full dependence–distance curves on each resample, so that the resulting variability reflects both the finite sample size and the residual serial dependence within storms. The composite-likelihood framework of Padoan et al. (2010) provides asymptotic standard errors but relies on correct parametric specification; the nonparametric summaries used here are instead assessed by block resampling and, in the simulation, by Monte Carlo evaluation against the known truth.

2.6 Seasonal and non-stationary extremal dependence

The seasonal question is a restricted form of non-stationary spatial extremal dependence: instead of one dependence structure for all daily maxima, our paper lets the dependence relation differ between two regimes, with season treated as a binary covariate. **huser2016** develop non-stationary dependence structures for spatial extremes, showing that extremal dependence itself, not only the marginal distribution, can vary with covariates. The seasonal split is a deliberately simple version of this idea and also addresses a stationarity concern: estimating one dependence curve from the unstratified daily maxima would mix large-scale winter storms with more localised summer convective events. Stratifying by season does not remove all non-stationarity, but it makes the working assumption more credible and gives each estimated curve a clearer interpretation.

In summary, Fisher and Tippett (1928), Gnedenko (1943), **balkema1974**, and **pickands1975** justify modelling tail observations and motivate the high-threshold analysis of daily maximum wind gusts. Haan and Ferreira (2006), Davison, Padoan, et al. (2012), and Padoan et al. (2010) provide the max-stable and pairwise-inference framework; Cooley et al. (2006), **cooley2012**, and Coles et al. (1999) motivate the pairwise estimands, with **wadsworth2012** and **huser2022** motivating the threshold-grid robustness check; the Brown–Resnick process (**brown1983**) with exact simulation (Dombry et al., 2016) supplies the benchmark for validation; and **huser2016** frames the seasonal comparison as non-stationary extremal dependence. The contribution of our paper is to combine these in a deliberately narrow setting: to establish, by simulation, when a seasonal difference in spatial extremal dependence can be recovered from finite daily maximum wind gusts on a fixed, irregular station network, and then to apply the validated estimators to Dutch wind gusts.

3 Methodology

This section defines the estimator used to compare spatial extremal dependence between winter and summer. The primary object is a season-specific Brown–Resnick max-stable model, fitted by pairwise composite likelihood, and the comparison is between its estimated dependence-range parameters ρ_W and ρ_S . The Brown–Resnick model is used as a parametric approximation to the spatial extremal dependence of daily maximum wind gusts within each season; the daily maxima are not assumed to be exactly max-stable. The nonparametric coefficients that are standard summaries of extremal dependence — the finite-level tail-dependence coefficient χ_u , the F -madogram and the empirical extremal coefficient — are retained in Section 3.7 as diagnostics for the fitted model rather than as estimands.

Let $X_{i,t}$ be the daily maximum wind gust at station $i = 1, \dots, N$ on day t , with $N = 33$ KNMI land and coastal stations, each observed on at least 95% of days over 1991–2026. Station i is at location $s_i \in \mathbb{R}^2$, and $d_{ij} = \|s_i - s_j\|$ is the great-circle distance between s_i and s_j in kilometres. On day t the network forms one spatial field

$$X_t = (X_{1,t}, \dots, X_{N,t}),$$

and the collection of such fields within a season is the input to estimation. The analysis is run separately for winter (DJF) and summer (JJA); $c \in \{W, S\}$ indexes the season and $\mathcal{T}_c$ the days in season c . December of year y is assigned to winter $y + 1$, so each winter is contiguous; incomplete boundary winters and the transitional seasons are dropped. For a pair (i, j) ,

$$\mathcal{T}_c^{ij} = \{t \in \mathcal{T}_c : X_{i,t} \text{ and } X_{j,t} \text{ both observed}\}, \quad n_{ij}^{(c)} = |\mathcal{T}_c^{ij}|,$$

and the pairwise likelihood contributions are computed on this pairwise-complete set.

3.1 Marginal standardisation

A max-stable model is specified on a common marginal scale, taken here to be unit Fréchet, $\Pr(Z \leq z) = \exp(-1/z)$ for $z > 0$. So that the seasonal comparison reflects spatial dependence and not differences in the local wind climate — coastal and inland stations differ in their gust distributions, and the two seasons differ as well — the margins are transformed separately by station *and* season. For station i and season c the empirical probability transform is

$$\hat{U}_{i,t}^{(c)} = \frac{1}{n_i^{(c)} + 1} \sum_{r \in \mathcal{T}_c} \mathbf{1}\{X_{i,r} \leq X_{i,t}\}, \quad t \in \mathcal{T}_c,$$

where $n_i^{(c)}$ is the number of observed daily maxima at station i in season c ; the $(n_i^{(c)} + 1)$ denominator keeps the transformed values inside $(0, 1)$. The standardised observation on the unit-Fréchet scale is

$$\hat{Z}_{i,t}^{(c)} = -\{\log \hat{U}_{i,t}^{(c)}\}^{-1}.$$

The rank transform avoids committing to a parametric marginal model. As a robustness alternative the margins are instead fitted by a generalised extreme value distribution $\hat{G}_i^{(c)}$ and mapped by $\hat{Z}_{i,t}^{(c)} = -1/\log \hat{G}_i^{(c)}(X_{i,t})$, which uses the extreme-value marginal limit but is sensitive to the marginal fit. Dependence estimation treats the standardised fields $\hat{Z}_t^{(c)}$ as the data.

3.2 Brown–Resnick model

For each season let $Z^{(c)} = \{Z^{(c)}(s) : s \in \mathbb{R}^2\}$ be a stationary max-stable process with unit-Fréchet margins. We take $Z^{(c)}$ to be a Brown–Resnick process (Brown and Resnick, 1977; Kabluchko et al., 2009): its extremal dependence is determined by the variogram of an underlying intrinsically stationary Gaussian process $W^{(c)}$ with $W^{(c)}(0) = 0$. Writing the variogram as the variance of increments,

$$\gamma_c(h) = \text{Var}\{W^{(c)}(s+h) - W^{(c)}(s)\},$$

the field is stationary and γ_c governs how fast extremal dependence decays with distance. We do not use the spectral representation directly; estimation uses only the bivariate distributions implied by γ_c , given in Section 3.4. The Brown–Resnick family is used as a parsimonious and interpretable parametric model for the season-specific dependence, not as a literal description of the data-generating mechanism.

3.3 Variogram and extremal coefficient

We use an isotropic powered-exponential variogram,

$$\gamma_c(d) = \left(\frac{d}{\rho_c}\right)^{\alpha_c}, \quad \rho_c > 0, \quad \alpha_c \in (0, 2],$$

so dependence depends on the locations only through the distance d . Here ρ_c is a range parameter: a larger ρ_c means slower decay of extremal dependence, hence a larger spatial footprint of extremes; α_c controls the smoothness of that decay. The pair $\vartheta_c = (\rho_c, \alpha_c)$ is the parameter of interest.

For two sites at distance d , the Brown–Resnick bivariate distribution function on the unit-Fréchet scale is of Hüsler–Reiss type,

$$\Pr\{Z^{(c)}(s_i) \leq z_i, Z^{(c)}(s_j) \leq z_j\} = \exp\{-V_d(z_i, z_j; \vartheta_c)\},$$

$$V_d(z_i, z_j; \vartheta_c) = \frac{1}{z_i} \Phi\left(\frac{a_c(d)}{2} + \frac{1}{a_c(d)} \log \frac{z_j}{z_i}\right) + \frac{1}{z_j} \Phi\left(\frac{a_c(d)}{2} + \frac{1}{a_c(d)} \log \frac{z_i}{z_j}\right),$$

where Φ is the standard normal distribution function and

$$a_c(d) = \sqrt{\gamma_c(d)} = \left(\frac{d}{\rho_c}\right)^{\alpha_c/2}.$$

The implied extremal coefficient is $\theta_c(d) = V_d(1, 1; \vartheta_c)$, that is

$$\theta_c(d) = 2\Phi\left(\frac{1}{2}a_c(d)\right) = 2\Phi\left(\frac{1}{2}\sqrt{\gamma_c(d)}\right) = 2\Phi\left(\frac{1}{2}(d/\rho_c)^{\alpha_c/2}\right) \in [1, 2],$$

decreasing from $\theta_c(0) = 1$ (complete dependence) to 2 (independence).

Convention. Throughout, γ_c is the variogram in the sense of the variance of increments. Some references state the same model through the semivariogram $\tilde{\gamma}_c = \gamma_c/2$, for which the extremal coefficient reads $\theta_c(d) = 2\Phi(\sqrt{\tilde{\gamma}_c(d)}/2)$; the two expressions are identical. The convention $a_c(d) = \sqrt{\gamma_c(d)}$, $\theta_c(d) = 2\Phi(a_c(d)/2)$ is used in the simulation of Section 4 and in the implementation.

3.4 Pairwise composite likelihood

Full-likelihood inference for max-stable processes is infeasible beyond a few sites, because the joint density involves a sum over all partitions of the locations (Padoan et al., 2010). We therefore estimate ϑ_c by pairwise composite likelihood, replacing the joint density by a product of bivariate densities. The bivariate density of a pair at distance d is obtained from V_d by differentiation,

$$f_d(z_i, z_j; \vartheta_c) = \left\{ \partial_{z_i} V_d \partial_{z_j} V_d - \partial_{z_i z_j}^2 V_d \right\} \exp\{-V_d(z_i, z_j; \vartheta_c)\},$$

whose closed form for the Hüsler–Reiss exponent is given by Padoan et al. (2010) and, for the Brown–Resnick process specifically, by Huser and Davison (2013). The pairwise composite log-likelihood for season c is

$$\ell_c(\vartheta_c) = \sum_{i < j} w_{ij} \sum_{t \in \mathcal{T}_c^{ij}} \log f_{d_{ij}}(\hat{Z}_{i,t}^{(c)}, \hat{Z}_{j,t}^{(c)}; \vartheta_c),$$

and the estimator is

$$\hat{\vartheta}_c = (\hat{\rho}_c, \hat{\alpha}_c) = \arg \max_{\vartheta_c \in \Theta} \ell_c(\vartheta_c), \quad \Theta = [\rho_L, \rho_U] \times [\alpha_L, 2].$$

The weights $w_{ij} \in \{0, 1\}$ restrict the sum to pairs with $d_{ij} \leq r$ for a cut-off r ; distant pairs are close to independent and carry little dependence information, and truncating them is the standard weighting in pairwise likelihood for spatial extremes (Padoan et al., 2010). The cut-off r , the parameter bounds, and the choice $w_{ij} \equiv 1$ versus a finite r are reported with the estimates and varied as a robustness check. As a further robustness alternative the pairwise likelihood may be censored below a high marginal threshold, so that only joint tail observations inform the fit (Huser and Davison, 2013); if this is used empirically, the simulation of Section 4 is re-run with the same censoring. Each day contributes one term per retained pair. Consecutive daily maxima within a multi-day storm are not independent; under stationarity and standard mixing conditions this serial dependence bears mainly on the sampling variability of $\hat{\vartheta}_c$, and hence on inference, which is handled by the bootstrap of Section 3.6.

3.5 Seasonal comparison and hypotheses

The model is estimated separately for the two seasons, giving $\hat{\vartheta}_W = (\hat{\rho}_W, \hat{\alpha}_W)$ and $\hat{\vartheta}_S = (\hat{\rho}_S, \hat{\alpha}_S)$; both the range and the smoothness are estimated freely in each season. The primary comparison is of the dependence ranges through

$$\hat{\Delta}_\rho = \hat{\rho}_W - \hat{\rho}_S.$$

Winter extremes are expected to have a larger spatial footprint than summer convective extremes, which under this model corresponds to a larger winter range. The hypotheses are

$$H_0 : \rho_W = \rho_S \quad \text{against} \quad H_1 : \rho_W > \rho_S.$$

Because winter and summer are estimated from disjoint days, $\hat{\rho}_W$ and $\hat{\rho}_S$ are computed on independent samples. The comparison is also read off the implied extremal-coefficient curves $\hat{\theta}_W(d)$ and $\hat{\theta}_S(d)$, which place the parameter difference on the dependence-versus-distance scale and are directly comparable to the diagnostics of Section 3.7. The smoothness parameters are compared secondarily through $\hat{\alpha}_W - \hat{\alpha}_S$; ρ is the object of interest.

3.6 Inference

Uncertainty is quantified primarily by a seasonal-year block bootstrap. The resampling unit is one season-year: within each season the season-years are resampled with replacement, independently across winter and summer, and for each replication the marginal transform of Section 3.1 and the estimator of Section 3.4 are recomputed, yielding $\hat{\vartheta}_W^*$, $\hat{\vartheta}_S^*$ and $\hat{\Delta}_\rho^*$. Resampling whole season-years rather than individual days preserves the within-storm serial dependence between consecutive daily maxima and the cross-station dependence among pairs that share a station, both of which a day-level resample would destroy. Percentile intervals for ρ_c , α_c and Δ_ρ , and pointwise bands for $\theta_c(d)$, are formed over B replications. The test of H_0 against H_1 is an interval-based one-sided bootstrap test: it rejects H_0 when the one-sided bootstrap percentile interval for $\hat{\Delta}_\rho$ excludes zero, equivalently when the lower percentile bound of $\hat{\Delta}_\rho$ lies above zero, with the corresponding one-sided bootstrap p -value reported. With on the order of 35 season-years per season the bands are correspondingly coarse, which the simulation of Section 4 quantifies.

As a parametric cross-check under the working assumption of independent replicates, composite-likelihood theory gives the Godambe (sandwich) information $G(\vartheta_c) = H(\vartheta_c) J(\vartheta_c)^{-1} H(\vartheta_c)$, where H is the expected negative Hessian of ℓ_c and J the variance of its score, and $\hat{\vartheta}_c$ is asymptotically normal with covariance $G(\vartheta_c)^{-1}$ (Padoan et al., 2010). Ordinary likelihood standard errors do not apply, because ℓ_c is a composite and not a genuine log-likelihood. Since the independence assumption underlying J is violated by serial dependence within storms, the sandwich errors are reported only alongside the block bootstrap, which is the basis for the reported intervals and the test.

3.7 Nonparametric diagnostics

The fitted model is assessed against three nonparametric pairwise summaries computed from the same standardised fields, none of which is the estimand. The first is the finite-level tail-dependence coefficient at level u ,

$$\hat{\chi}_{u,ij}^{(c)} = \frac{\#\{t \in \mathcal{T}_c^{ij} : \hat{U}_{i,t}^{(c)} > u, \hat{U}_{j,t}^{(c)} > u\}}{\#\{t \in \mathcal{T}_c^{ij} : \hat{U}_{i,t}^{(c)} > u\}},$$

the empirical probability that station j is extreme given that station i is, reported on the grid $u \in \{0.95, 0.975, 0.99\}$ and symmetrised over the two conditioning directions (Coles et al., 1999). The second is the F -madogram and its extremal coefficient (Cooley et al., 2006),

$$\hat{\nu}_{F,ij}^{(c)} = \frac{1}{2n_{ij}^{(c)}} \sum_{t \in \mathcal{T}_c^{ij}} |\hat{U}_{i,t}^{(c)} - \hat{U}_{j,t}^{(c)}|, \quad \hat{\theta}_{ij}^{(c)} = \frac{1 + 2\hat{\nu}_{F,ij}^{(c)}}{1 - 2\hat{\nu}_{F,ij}^{(c)}} \in [1, 2].$$

Under a Brown–Resnick model the implied finite-level coefficient is $\chi_u^{\text{BR}}(d) = (1 - 2u + u^{\theta_c(d)})/(1 - u)$, and the implied extremal coefficient is $\theta_c(d)$. The diagnostics check whether the fitted $\hat{\theta}_c(\cdot)$ and the implied $\chi_u^{\text{BR}}(\cdot)$ track the pairwise estimates $(d_{ij}, \hat{\theta}_{ij}^{(c)})$ and $(d_{ij}, \hat{\chi}_{u,ij}^{(c)})$ across distance. Systematic departure —

for instance pairwise $\hat{\chi}_{u,ij}^{(c)}$ decaying towards zero at the highest level while the fitted curve stays bounded away from it — indicates that the asymptotic-dependence assumption of the Brown–Resnick model is questionable for these data (Davison, Huser, et al., 2013). Pairwise estimates and the fitted curve are always shown together, so the comparison is visible rather than smoothed away. The diagnostics use scalar distance and therefore assume isotropy; departures associated with prevailing storm direction are examined as a robustness check.

The finite-sample behaviour of $\hat{\vartheta}_c$ and of the test of H_0 is evaluated in Section 4.

4 Simulation Study

4.1 Purpose

The empirical analysis estimates $\vartheta_W = (\rho_W, \alpha_W)$ and $\vartheta_S = (\rho_S, \alpha_S)$ from a single, seasonally split sample of daily maxima on an irregular station network, and reads the winter–summer comparison from $\widehat{\Delta}_\rho = \widehat{\rho}_W - \widehat{\rho}_S$. The simulation asks whether the estimator of Section 3.4 recovers the parameters, and whether the interval-based one-sided bootstrap test of $H_0 : \rho_W = \rho_S$ against $H_1 : \rho_W > \rho_S$ has the intended size and power, at a sample size and geometry matching the application. The data are generated from Brown–Resnick processes for which ϑ_c and the implied curves $\theta_c(d)$ are known; this is a controlled truth, not a claim that Dutch gusts are Brown–Resnick. Three quantities are evaluated: bias and root mean squared error of $\widehat{\rho}_c$ and $\widehat{\alpha}_c$, coverage of the bootstrap intervals, and the size and power of the test of H_0 .

4.2 Spatial design

Fields are simulated on the empirical station set, so the simulated and observed inter-station distances coincide. Let $s_1, \dots, s_N$ be the $N = 33$ retained station locations and $d_{ij} = \|s_i - s_j\|$ the great-circle distance in kilometres. Each simulated field is one realisation of the spatial process, the analogue of one day; n fields are drawn per season, with n matching the per-season count of daily maxima in the data. Sample size is treated as a design factor, with n reduced below its empirical value to trace how the estimator degrades as the number of fields shrinks.

4.3 Brown–Resnick data-generating process

The data-generating process is the Brown–Resnick max-stable process (Brown and Resnick, 1977; Kabluchko et al., 2009) with unit-Fréchet margins and variogram

$$\gamma_c(d) = \left(\frac{d}{\rho_c}\right)^{\alpha_c}, \quad \rho_c > 0, \alpha_c \in (0, 2],$$

matching the model of Section 3.3. Fields are drawn by the exact algorithm of Dombry et al. (2016), so the realised dependence equals the target rather than an approximation to it. The implied extremal coefficient is $\theta_c(d) = 2\Phi\left(\frac{1}{2}\sqrt{\gamma_c(d)}\right)$, with Φ the standard normal distribution function, and the implied finite-level coefficient at level u is $\chi_u^{(c)}(d) = (1 - 2u + u^{\theta_c(d)})/(1 - u)$; both serve as known targets for the diagnostics of Section 3.7.

4.4 Null and alternative designs

Two regimes, labelled winter and summer, mirror the empirical comparison. Under the null the regimes share the variogram, $\rho_W = \rho_S$, so there is no seasonal difference; under the alternative the winter range exceeds the summer range, $\rho_W > \rho_S$, which implies $\theta_W(d) < \theta_S(d)$ for $d > 0$: winter extremes are more strongly dependent and remain so over longer distances. In both designs the data-generating smoothness is common to the two regimes, $\alpha_W = \alpha_S$, so the alternative isolates a difference in range. Each regime is then estimated with α free, matching the empirical model of Section 3.5; a simplified design that imposes a single common α in estimation is run as well, since fixing the shared smoothness can sharpen the comparison of ρ . The parameter values are chosen so that the two curves are distinguishable over the distances the network spans, and are reported with the results; they are not empirical estimates.

4.5 Estimation in each replication

Each Monte Carlo replication applies the empirical pipeline unchanged. One winter and one summer sample of n fields are drawn. Within each regime the margins are transformed as in Section 3.1; on exact unit-Fréchet fields this step is close to the identity, so applying it keeps the marginal estimation error of the empirical procedure in the simulation rather than removing it. The parameters are estimated by the pairwise composite likelihood of Section 3.4, giving $\widehat{\vartheta}_W$, $\widehat{\vartheta}_S$ and $\widehat{\Delta}_\rho$, and the seasonal-year block bootstrap of Section 3.6 is run within the replication by arranging the n fields into pseudo-season-years and resampling them. Pointwise targets and the test decision use the same rules as the application. The nonparametric F -madogram and χ_u coefficients (Cooley et al., 2006) are also computed and compared

Algorithm 1 Monte Carlo evaluation of the seasonal Brown–Resnick estimator

Require: locations $\{s_i\}_{i=1}^N$; per-season size n ; ranges ρ_W, ρ_S and smoothness α ; replications R ; bootstrap size B ; parameter space Θ

```

1: for  $r = 1, \dots, R$  do
2:   for  $c \in \{W, S\}$  do
3:     draw  $n$  exact unit-Fréchet Brown–Resnick fields with variogram  $\gamma_c(d) = (d/\rho_c)^\alpha$ 
4:     transform margins by station to the unit-Fréchet scale (Section 3.1)
5:      $\hat{\vartheta}_c \leftarrow \arg \max_{\vartheta \in \Theta} \ell_c(\vartheta)$  by pairwise composite likelihood (Section 3.4)
6:   end for
7:    $\hat{\Delta}_\rho \leftarrow \hat{\rho}_W - \hat{\rho}_S$ 
8:   for  $b = 1, \dots, B$  do
9:     arrange each regime’s  $n$  fields into pseudo-season-years; resample within regime
10:    recompute  $\hat{\vartheta}_W^*, \hat{\vartheta}_S^*, \hat{\Delta}_\rho^*$ 
11:  end for
12:  store  $\hat{\vartheta}_W, \hat{\vartheta}_S, \hat{\Delta}_\rho$ , their percentile intervals, and the interval-based one-sided test decision for  $H_0$ 
13: end for

```

Ensure: across r : bias and RMSE of $\hat{\rho}_c, \hat{\alpha}_c, \hat{\Delta}_\rho$; interval coverage; size (null) and power (alternative)

Table 1: Simulation results, to be completed after estimation. Monte Carlo bias and RMSE of the Brown–Resnick estimates and of $\hat{\Delta}_\rho$, and bootstrap interval coverage, under the null and alternative designs at per-season size n .

Design	Parameter	True	Bias	RMSE	Coverage
Null	ρ_W	–	–	–	–
	ρ_S	–	–	–	–
	α_W	–	–	–	–
	α_S	–	–	–	–
	Δ_ρ	0	–	–	–
Alt.	ρ_W	–	–	–	–
	ρ_S	–	–	–	–
	α_W	–	–	–	–
	α_S	–	–	–	–
	Δ_ρ	–	–	–	–

to their known curves, as a secondary check on the diagnostics. Algorithm 1 collects the per-replication procedure.

4.6 Performance measures

Let $\mathbb{E}_{\text{MC}}$ denote the average over the R replications. For each season-specific parameter $\beta \in \{\rho_c, \alpha_c\}$ and for Δ_ρ the simulation reports

$$\text{Bias}(\beta) = \mathbb{E}_{\text{MC}}[\hat{\beta}] - \beta, \quad \text{RMSE}(\beta) = \{\mathbb{E}_{\text{MC}}[(\hat{\beta} - \beta)^2]\}^{1/2}.$$

Interval coverage is the proportion of replications whose bootstrap percentile interval for β contains the true value. For the interval-based one-sided bootstrap test of $H_0 : \rho_W = \rho_S$ against $H_1 : \rho_W > \rho_S$ (Section 3.6), size is the rejection rate under the null design and should sit near the nominal level, and power is the rejection rate under the alternative. Curve recovery is summarised by the bias and RMSE of the implied extremal coefficient $\hat{\theta}_c(d_0)$ at reference distances d_0 , evaluated against $\theta_c(d_0)$. Because the fields are drawn independently across days, the coverage and size figures are for the favourable case of no residual temporal dependence; this is restated in Section 4.7. Table 1 collects the parameter results and Table 2 the size and power.

4.7 Interpretation

The simulation fixes what the empirical comparison can support. Small bias and near-nominal coverage at the empirical sample size would mean $\hat{\Delta}_\rho$ and its interval can be read directly; the power results map

Table 2: Size and power of the interval-based one-sided bootstrap test of $H_0 : \rho_W = \rho_S$ against $H_1 : \rho_W > \rho_S$, to be completed after estimation, by per-season size n and nominal level. Size is the rejection rate under the null design; power under the alternative.

n	Nominal level	Size (null)	Power (alt.)
—	0.05	—	—
—	0.05	—	—
—	0.05	—	—

the smallest difference in dependence range that is detectable at this sample size and geometry, which is the relevant quantity for an irregular network with a short tail sample. Two limits are expected in the numbers. The range ρ is typically estimated less precisely than α , and precision in ρ deteriorates once the network’s maximum distance is small relative to ρ , since the extremal coefficient is then close to its small-distance values across all pairs and carries little information about the range. And because the simulated fields are independent across days, the coverage and size figures are an upper bound on what the seasonal-year bootstrap attains on the data, where consecutive daily maxima within a storm remain dependent; closing that gap would require simulating temporally dependent daily maxima, which is not needed to evaluate recovery of a known spatial difference and is left as an extension.

A Mathematical details for pairwise extremal dependence estimators

A.1 Probability integral transform

We prove that the marginal transform $U_{i,t} = F_i(X_{i,t})$ has a standard uniform distribution when F_i is continuous.

Proof. Let $X_{i,t}$ be a real-valued random variable with continuous distribution function F_i . Define

$$U_{i,t} = F_i(X_{i,t}).$$

For $u < 0$,

$$\Pr(U_{i,t} \leq u) = 0.$$

For $u > 1$,

$$\Pr(U_{i,t} \leq u) = 1.$$

Let $u \in [0, 1]$. Define the generalized inverse

$$F_i^{-1}(u) = \inf\{x \in \mathbb{R} : F_i(x) \geq u\}.$$

Since F_i is continuous,

$$F_i(F_i^{-1}(u)) = u.$$

Then

$$\begin{aligned} \Pr(U_{i,t} \leq u) &= \Pr(F_i(X_{i,t}) \leq u) \\ &= \Pr(X_{i,t} \leq F_i^{-1}(u)) \\ &= F_i(F_i^{-1}(u)) \\ &= u. \end{aligned}$$

Hence

$$\Pr(U_{i,t} \leq u) = u, \quad u \in [0, 1].$$

Therefore

$$U_{i,t} \sim \text{Uniform}(0, 1).$$

For two stations i and j , define

$$U_{i,t} = F_i(X_{i,t}), \quad U_{j,t} = F_j(X_{j,t}).$$

Then

$$U_{i,t} \sim \text{Uniform}(0, 1), \quad U_{j,t} \sim \text{Uniform}(0, 1).$$

The marginal distributions are therefore equal after transformation, and the joint distribution of $(U_{i,t}, U_{j,t})$ contains only dependence information. $\square$

A.2 Relation between χ_{ij} and θ_{ij} under max-stability

We prove that, under bivariate max-stability with standard uniform margins, the upper tail dependence coefficient and the extremal coefficient satisfy $\chi_{ij} = 2 - \theta_{ij}$.

Proof. Let (U_i, U_j) have standard uniform margins. Assume the bivariate extreme-value copula representation

$$\Pr(U_i \leq u, U_j \leq u) = u^{\theta_{ij}}, \quad u \in (0, 1),$$

where

$$\theta_{ij} \in [1, 2].$$

The upper tail dependence coefficient is

$$\chi_{ij} = \lim_{u \uparrow 1} \Pr(U_j > u \mid U_i > u).$$

By the definition of conditional probability,

$$\Pr(U_j > u \mid U_i > u) = \frac{\Pr(U_i > u, U_j > u)}{\Pr(U_i > u)}.$$

Since $U_i \sim \text{Uniform}(0, 1)$,

$$\Pr(U_i > u) = 1 - u.$$

By inclusion-exclusion,

$$\begin{aligned} \Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \leq u) - \Pr(U_j \leq u) + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - u - u + u^{\theta_{ij}} \\ &= 1 - 2u + u^{\theta_{ij}}. \end{aligned}$$

Therefore

$$\Pr(U_j > u \mid U_i > u) = \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Thus

$$\chi_{ij} = \lim_{u \uparrow 1} \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Both numerator and denominator converge to zero:

$$\lim_{u \uparrow 1} (1 - 2u + u^{\theta_{ij}}) = 1 - 2 + 1 = 0,$$

$$\lim_{u \uparrow 1} (1 - u) = 0.$$

By l'Hopital's rule,

$$\begin{aligned} \chi_{ij} &= \lim_{u \uparrow 1} \frac{-2 + \theta_{ij} u^{\theta_{ij}-1}}{-1} \\ &= \frac{-2 + \theta_{ij}}{-1} \\ &= 2 - \theta_{ij}. \end{aligned}$$

Hence

$$\boxed{\chi_{ij} = 2 - \theta_{ij}.$$

The endpoints are consistent with the interpretation:

$$\theta_{ij} = 1 \implies \chi_{ij} = 1,$$

$$\theta_{ij} = 2 \implies \chi_{ij} = 0.$$

□

A.3 Extremal coefficient as the exponent measure evaluated at $(1, 1)$

We prove that, for a bivariate max-stable distribution with unit Fréchet margins, the pairwise extremal coefficient is $\theta_{ij} = V_{ij}(1, 1)$.

Proof. Let (Z_i, Z_j) denote the bivariate vector

$$(Z_i, Z_j) = (Z(s_i), Z(s_j)).$$

Assume unit Fréchet margins:

$$\Pr(Z_i \leq z) = \exp(-1/z), \quad z > 0,$$

$$\Pr(Z_j \leq z) = \exp(-1/z), \quad z > 0.$$

Let the joint distribution have exponent measure representation

$$\Pr(Z_i \leq z_i, Z_j \leq z_j) = \exp\{-V_{ij}(z_i, z_j)\}, \quad z_i, z_j > 0.$$

Set

$$z_i = z, \quad z_j = z.$$

Then

$$\Pr(Z_i \leq z, Z_j \leq z) = \exp\{-V_{ij}(z, z)\}.$$

For a max-stable distribution with unit Fréchet margins, the exponent measure is homogeneous of order -1 :

$$V_{ij}(az_i, az_j) = a^{-1}V_{ij}(z_i, z_j), \quad a > 0.$$

Take

$$a = z, \quad z_i = 1, \quad z_j = 1.$$

Then

$$\begin{aligned} V_{ij}(z, z) &= V_{ij}(z \cdot 1, z \cdot 1) \\ &= z^{-1}V_{ij}(1, 1). \end{aligned}$$

Therefore

$$\begin{aligned} \Pr(Z_i \leq z, Z_j \leq z) &= \exp\{-z^{-1}V_{ij}(1, 1)\} \\ &= \exp\left\{-\frac{V_{ij}(1, 1)}{z}\right\}. \end{aligned}$$

The pairwise extremal coefficient θ_{ij} is defined by

$$\Pr(Z_i \leq z, Z_j \leq z) = \Pr(Z_i \leq z)^{\theta_{ij}}.$$

Since

$$\Pr(Z_i \leq z) = \exp(-1/z),$$

we have

$$\begin{aligned} \Pr(Z_i \leq z)^{\theta_{ij}} &= [\exp(-1/z)]^{\theta_{ij}} \\ &= \exp(-\theta_{ij}/z). \end{aligned}$$

Equating the two expressions for the same joint probability gives

$$\exp\left\{-\frac{V_{ij}(1, 1)}{z}\right\} = \exp\left\{-\frac{\theta_{ij}}{z}\right\}.$$

Taking logarithms gives

$$-\frac{V_{ij}(1, 1)}{z} = -\frac{\theta_{ij}}{z}.$$

Multiplying by $-z$ gives

$$V_{ij}(1, 1) = \theta_{ij}.$$

Hence

$$\boxed{\theta_{ij} = V_{ij}(1, 1)}.$$

The bounds follow from the two limiting cases. Perfect dependence gives

$$Z_i = Z_j,$$

so

$$\Pr(Z_i \leq z, Z_j \leq z) = \Pr(Z_i \leq z) = \exp(-1/z),$$

which implies

$$\theta_{ij} = 1.$$

Independence gives

$$\begin{aligned}\Pr(Z_i \leq z, Z_j \leq z) &= \Pr(Z_i \leq z) \Pr(Z_j \leq z) \\ &= \exp(-1/z) \exp(-1/z) \\ &= \exp(-2/z),\end{aligned}$$

which implies

$$\theta_{ij} = 2.$$

Therefore

$$1 \leq \theta_{ij} \leq 2.$$

□

A.4 F-madogram relation to the extremal coefficient

We prove that, under bivariate max-stability with standard uniform margins, the F-madogram satisfies $\theta_{ij} = (1 + 2\nu_F)/(1 - 2\nu_F)$.

Proof. Let

$$U_i = F_i(Z_i), \quad U_j = F_j(Z_j),$$

where U_i and U_j have standard uniform margins. The F-madogram is

$$\nu_F(i, j) = \frac{1}{2} \mathbb{E}|U_i - U_j|.$$

Use the identity

$$|a - b| = a + b - 2 \min(a, b), \quad a, b \in [0, 1].$$

Then

$$\begin{aligned}\mathbb{E}|U_i - U_j| &= \mathbb{E}(U_i + U_j - 2 \min(U_i, U_j)) \\ &= \mathbb{E}U_i + \mathbb{E}U_j - 2\mathbb{E} \min(U_i, U_j).\end{aligned}$$

Since $U_i, U_j \sim \text{Uniform}(0, 1)$,

$$\mathbb{E}U_i = \frac{1}{2}, \quad \mathbb{E}U_j = \frac{1}{2}.$$

Therefore

$$\mathbb{E}|U_i - U_j| = 1 - 2\mathbb{E} \min(U_i, U_j).$$

Thus

$$\nu_F(i, j) = \frac{1}{2} - \mathbb{E} \min(U_i, U_j).$$

It remains to compute $\mathbb{E} \min(U_i, U_j)$. For any random variable $Y \in [0, 1]$,

$$\mathbb{E}Y = \int_0^1 \Pr(Y > u) du.$$

Set

$$Y = \min(U_i, U_j).$$

Then

$$\begin{aligned}\mathbb{E} \min(U_i, U_j) &= \int_0^1 \Pr(\min(U_i, U_j) > u) du \\ &= \int_0^1 \Pr(U_i > u, U_j > u) du.\end{aligned}$$

By inclusion-exclusion,

$$\begin{aligned}\Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \leq u) - \Pr(U_j \leq u) + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - u - u + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - 2u + \Pr(U_i \leq u, U_j \leq u).\end{aligned}$$

Under bivariate max-stability with extremal coefficient θ_{ij} ,

$$\Pr(U_i \leq u, U_j \leq u) = u^{\theta_{ij}}.$$

Therefore

$$\begin{aligned}\mathbb{E} \min(U_i, U_j) &= \int_0^1 \{1 - 2u + u^{\theta_{ij}}\} du \\ &= \int_0^1 1 du - 2 \int_0^1 u du + \int_0^1 u^{\theta_{ij}} du \\ &= 1 - 2 \left[\frac{u^2}{2} \right]_0^1 + \left[\frac{u^{\theta_{ij}+1}}{\theta_{ij}+1} \right]_0^1 \\ &= 1 - 1 + \frac{1}{\theta_{ij}+1} \\ &= \frac{1}{\theta_{ij}+1}.\end{aligned}$$

Hence

$$\begin{aligned}\nu_F(i, j) &= \frac{1}{2} - \frac{1}{\theta_{ij}+1} \\ &= \frac{\theta_{ij}+1}{2(\theta_{ij}+1)} - \frac{2}{2(\theta_{ij}+1)} \\ &= \frac{\theta_{ij}-1}{2(\theta_{ij}+1)}.\end{aligned}$$

Thus

$$2\nu_F(i, j) = \frac{\theta_{ij}-1}{\theta_{ij}+1}.$$

Solve for θ_{ij} :

$$\begin{aligned}2\nu_F(\theta_{ij}+1) &= \theta_{ij}-1, \\ 2\nu_F\theta_{ij} + 2\nu_F &= \theta_{ij}-1, \\ \theta_{ij} - 2\nu_F\theta_{ij} &= 1 + 2\nu_F, \\ \theta_{ij}(1 - 2\nu_F) &= 1 + 2\nu_F, \\ \theta_{ij} &= \frac{1 + 2\nu_F}{1 - 2\nu_F}.\end{aligned}$$

Hence

$$\boxed{\theta_{ij} = \frac{1 + 2\nu_F(i, j)}{1 - 2\nu_F(i, j)}}.$$

Replacing $\nu_F(i, j)$ by the empirical F-madogram

$$\hat{\nu}_F(i, j; c) = \frac{1}{2B^{(c)}} \sum_{b=1}^{B^{(c)}} \left| \hat{F}_i^{(c)}(M_{i,b}^{(c)}) - \hat{F}_j^{(c)}(M_{j,b}^{(c)}) \right|$$

gives the estimator

$$\hat{\theta}_{ij}^{(c)} = \frac{1 + 2\hat{\nu}_F(i, j; c)}{1 - 2\hat{\nu}_F(i, j; c)}.$$

□

A.5 Consistency of the empirical finite-level tail-dependence estimator

We prove that, under within-season stationarity and ergodicity, the empirical estimator $\widehat{\chi}_{ij}^{(c)}(u)$ converges to the finite-level conditional tail probability $\Pr(U_j^{(c)} > u \mid U_i^{(c)} > u)$.

Proof. Fix a season

$$c \in \{W, S\}.$$

Fix a pair of stations

$$(i, j).$$

Let

$$\{(U_{i,t}^{(c)}, U_{j,t}^{(c)}) : t \in T_c\}$$

be the season- c transformed pair process. Assume this process is strictly stationary and ergodic. Fix

$$u \in (0, 1).$$

Define

$$A_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\},$$

and

$$B_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u\}.$$

Since $(U_{i,t}^{(c)}, U_{j,t}^{(c)})$ is strictly stationary and ergodic, any measurable transformation of it is also strictly stationary and ergodic. Therefore

$$\{A_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic, and

$$\{B_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic. Moreover,

$$0 \leq A_t^{(c)}(u) \leq 1, \quad 0 \leq B_t^{(c)}(u) \leq 1.$$

Hence

$$\mathbb{E}|A_t^{(c)}(u)| < \infty, \quad \mathbb{E}|B_t^{(c)}(u)| < \infty.$$

By the ergodic theorem,

$$\frac{1}{|T_c|} \sum_{t \in T_c} A_t^{(c)}(u) \xrightarrow{p} \mathbb{E}A_t^{(c)}(u),$$

and

$$\frac{1}{|T_c|} \sum_{t \in T_c} B_t^{(c)}(u) \xrightarrow{p} \mathbb{E}B_t^{(c)}(u).$$

By the definition of $A_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}A_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u). \end{aligned}$$

By the definition of $B_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}B_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u). \end{aligned}$$

Since $U_{i,t}^{(c)}$ has a standard uniform margin,

$$\Pr(U_{i,t}^{(c)} > u) = 1 - u.$$

For $u \in (0, 1)$,

$$1 - u > 0.$$

The empirical finite-level tail-dependence estimator is

$$\hat{\chi}_{ij}^{(c)}(u) = \frac{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\}}{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u\}}.$$

Using $A_t^{(c)}(u)$ and $B_t^{(c)}(u)$,

$$\begin{aligned} \hat{\chi}_{ij}^{(c)}(u) &= \frac{\sum_{t \in T_c} A_t^{(c)}(u)}{\sum_{t \in T_c} B_t^{(c)}(u)} \\ &= \frac{|T_c|^{-1} \sum_{t \in T_c} A_t^{(c)}(u)}{|T_c|^{-1} \sum_{t \in T_c} B_t^{(c)}(u)}. \end{aligned}$$

By the continuous mapping theorem,

$$\begin{aligned} \hat{\chi}_{ij}^{(c)}(u) &\xrightarrow{p} \frac{\Pr(U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u)}{\Pr(U_{i,t}^{(c)} > u)} \\ &= \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u). \end{aligned}$$

Therefore

$$\boxed{\hat{\chi}_{ij}^{(c)}(u) \xrightarrow{p} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)}$$

for every fixed $u \in (0, 1)$. If the limit

$$\chi_{ij}^{(c)} = \lim_{u \uparrow 1} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)$$

exists, then the finite-level probability estimated above is the finite- u approximation to the asymptotic tail-dependence coefficient. $\square$

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Table 3: All available variables in hourly-in-situ-meteorological-observations.

Param.	Description	Unit	Observed property	Com.
DD	Past 10 min mean wind direction at 10m (360=N, 90=E, 180=S, 270=W, 0=calm, 990=var)	°	Wind from direction	
DR	Hourly precipitation duration	h	Duration of precipitation	
EE	Hourly mean vapour pressure	hPa	Water vapour partial pressure	
FF	Past 10 min mean wind speed at 10m	m/s	Wind speed	
FH	Hourly mean wind speed at 10m	m/s	Wind speed	
FX	Hourly maximum wind gust at 10m	m/s	Wind speed of gust	
IX	Hourly indicator present weather code (1-3: manned; 4-6: auto; 7: auto-set)	–	Indicator present weather code	
N	Past 10 min cloud cover (in okta; 9 = sky invisible)	okta	Cloud area fraction	
P	Past 1 min mean air pressure at sea level	hPa	Air pressure at mean sea level	
Q	Hourly global solar radiation	J/cm ²	Downwelling shortwave flux	
RH	Hourly precipitation amount	mm	Precipitation amount	¹
SQ	Hourly sunshine duration, calculated from global solar radiation	h	Duration of sunshine	²
T	Past 1 min air temperature at 1.50m	°C	Air temperature	
T10N	Past 6 hour minimum air temperature at 10cm	°C	Air temperature	
TD	Past 1 min dew point temperature at 1.50m	°C	Dew point temperature	
U	Past 1 min relative atmospheric humidity at 1.50m	%	Relative humidity	
VV	Past 10 min horizontal visibility (0: <100m; 1-49: 100m-5km; 50-89: 5km to >70km)	–	Visibility in air	
W1	Hourly indicator of fog (0 = no, 1 = occurred)	–	Fog	
W2	Hourly indicator of rainfall (0 = no, 1 = occurred)	–	Rain	
W3	Hourly indicator of snow (0 = no, 1 = occurred)	–	Snow	
W5	Hourly indicator of thunder (0 = no, 1 = occurred)	–	Thunder	
W6	Hourly indicator of ice formation (0 = no, 1 = occurred)	–	Ice formation	
WW	Hourly present weather code, aggregated from preceding 6 hours (WMO table 4680)	–	Present weather code	

Notes: ¹ Precipitation detection minimum is 0.05 mm. ² Sunshine duration measured minimum is 0.05 h.